Robinson Vidva

Computational Biologist

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SUMMARY

Computational biologist and bioinformatics scientist with more than 16 years of experience across systems biology modeling, multi-omics analysis, machine learning, and biostatistics for computational oncology, immuno-oncology, drug discovery, and pediatric neurodevelopment. Twelve years building and leading precision-oncology and immunology modeling platforms at Cellworks, three years running an independent consultancy, and, since 2025, bioinformatics and biostatistics for a neonatal neuroscience laboratory at Children's National Hospital. Integrates mutational, gene-expression, imaging, and clinical data to model treatment response and identify biomarkers. Managed a team of 12 scientists. First author of one and co-author of six further peer-reviewed journal articles, two preprints, and 14 conference abstracts (150+ citations); named inventor on two patented method-of-use drug combinations (three grants, US and European) and three published patent applications; developer of open-source research tools, including a published lab-animal colony management platform.

TECHNICAL SKILLS

Computational biology and bioinformatics: next-generation sequencing (NGS) analysis; bulk RNA-seq, genomics, metabolomics; variant analysis (SNVs, CNVs); differential expression, pathway, gene-set enrichment, and cell-cell communication analysis

Systems biology, statistics, and machine learning: mechanistic network, ODE, and stochastic pathway models; patient-specific models of drug response; generalized linear, mixed-effects, segmented, logistic, modified Poisson, and survival models; scikit-learn, TensorFlow, Keras, PyTorch

Programming and infrastructure: Python, R, SQL, MATLAB, C/C++, Unix shell, JavaScript/TypeScript, PHP; Git, CI/CD, Docker and Singularity, Snakemake, LSF, AWS; database design; full-stack web development

Clinical and multimodal data: quality control and data-issue review of de-identified clinical datasets covering electronic health records (EHR), imaging (MRI, ultrasound), laboratory, and neurodevelopmental data; HIPAA and protected health information (PHI) handling; data use agreements

Behavioral phenotyping: DeepLabCut, B-SOiD, BONSAI, OpenCV, MediaPipe; three-chamber sociability and Erasmus Ladder analysis

EXPERIENCE

Research Technician (Bioinformatics)

Children's National Hospital (Kratimenos Lab), Washington, DC | 03/2025 - Present

- Quality-check and integrate the de-identified clinical datasets (electronic health record (EHR), imaging, laboratory, metabolomic, and transcriptomic data) for two preterm-infant studies (178 and 181 infants).
- Run the statistical analyses relating clinical exposures to brain development, including more than 1,000 serial cranial ultrasound scans, using linear, mixed-effects, segmented, logistic, and modified Poisson models, with stratification and sensitivity analyses.
- Build and maintain reproducible RNA-seq pipelines for mouse and human cerebellar studies of prematurity insults, necrotizing enterocolitis, and sudden infant death syndrome; run differential expression, pathway, and Gene Ontology (GO) enrichment analyses; provide RNA-seq analysis for an astrocytoma immunotherapy study run by a collaborating laboratory.
- Quantify rodent behavior with DeepLabCut and B-SOiD in a mouse model of prematurity insults and its early rehabilitation extension; develop statistical and machine-learning models on Erasmus Ladder data from 160 preterm mice.
- Prepare figures, tables, and sequencing-data repository submissions for the laboratory's manuscripts and check them against the deposited data; co-authored a review on immune modulation in skin cancer, contributing the cancer-immunology content and figures (Frontiers in Immunology, 2025).

Founder and Lead Computational Biologist

Digirobi Solutions, Bengaluru, India | 09/2019 - 06/2022

- Ran an independent computational biology consultancy for commercial and academic clients in cancer therapeutics and research software.
- Assembled cancer genomics, drug, and nutrition databases for a nutrigenomics platform, and co-invented a cancer combination therapy later patented in the US and Europe.
- Developed a mechanistic model of Src kinase signaling in neonatal hypoxic neuronal injury (MATLAB SimBiology), which resulted in a peer-reviewed paper (2022); designed and built MyVivarium, an open-source colony-management platform with IoT sensing (published 2025).

Senior Lead Scientist

Cellworks Life, Bengaluru, India | 07/2007 - 09/2019

- Promoted through Associate Biomodeling Scientist (2007), Associate Lead Scientist (2008), Lead Scientist (2014), and Senior Lead Scientist (2017); mentored more than 25 junior scientists and delivered more than 15 internal training sessions.
- Led a team of 12 scientists developing computational precision-medicine platforms for oncology, and set the standards for model reproducibility and validation.
- Using TCGA, MSK-IMPACT, and proprietary genomic, transcriptomic, and pharmacological data, designed and validated patient-specific drug-response models across solid and hematologic cancers; in the Beat AML study, predicted sensitivity or resistance to a BET inhibitor agreed with the ex vivo IC50 classification for 93 of 100 acute myeloid leukemia (AML) patients (Leukemia Research, 2019); the same approach was applied to venetoclax, dasatinib, CDK4/6 and IDH inhibitors, and pediatric AML chemotherapy response (Blood, 2016 - 2018).
- Assembled a cell-line resource (more than 500 lines with IC50 profiles), computational drug models for about 200 compounds, and more than 15 curated clinical datasets used to train, validate, and benchmark the models.
- Co-invented the PTEN/TP53-targeted cancer combination therapy (US 10,098,880 B2), drafting the claims and handling prosecution through examiner objections; contributed to two further patent applications.
- Built and validated an eight-cell-type systems biology model of rheumatoid arthritis against in vivo data and identified two synergistic repurposed drug combinations, one of which reduced disease progression in a murine arthritis model.

SELECTED TECHNICAL PROJECTS

- **MyVivarium:** cloud platform for lab-animal colony management with IoT sensing (PHP, MySQL, JavaScript); CSBJ 2025; in use at three laboratories; version 2 being deployed at the University of Dayton. Source: github.com/myvivarium/MyVivarium
- **neokine:** client-side MediaPipe pose visualization with joint-angle, velocity, and asymmetry plots. robinsonvidva.com/neokine
- **drugtargets:** drug-target exploration and repurposing tool on Open Targets and openFDA. drugtargets.robinsonvidva.com

Additional open-source tools: enrichlite (browser-based gene-set over-representation analysis) and DrugInteract (client-side drug-interaction reference). Details at robinsonvidva.com/tools

EDUCATION

M.S., Data Analytics, Western Governors University, Salt Lake City, UT, 2025

M.B.A., Project Management, Sikkim Manipal University, India, 2020

B.Tech. (Bachelor of Technology), Biotechnology, Bharathidasan University, India; First Class with Distinction, 2007

PATENTS

- EP 3,861,992 B1 (2023) and US 11,478,444 B2 (2022), granted: plant-derived supplements combined with antineoplastic compounds for hematological malignancies.
- US 10,098,880 B2 (2018), granted: nelfinavir, metformin, and rosuvastatin combination for PTEN/TP53-aberrant cancers.
- US 2015/0302167 A1, US 2015/0098993 A1, US 2014/0127295 A1, published applications: therapeutic-development method; compositions for inflammatory disease.

SELECTED PUBLICATIONS

- Giannopoulos K, Karikis I, Byrd C, Sanidas G, Wolff N, Triantafyllou M, Simonti G, Vidva R, et al. Eph/Ephrin-mediated immune modulation: a potential therapeutic target for skin cancer. Front Immunol. 2025;16:1539567.
- Vidva R, Raza MA, Prabhakaran J, Sheikh A, Sharp A, Ott H, et al. MyVivarium: a cloud-based lab animal colony management application with real-time ambient sensing. Comput Struct Biotechnol J. 2025;27:612-23.
- Kratimenos P, Vij A, Vidva R, Koutroulis I, Delivoria-Papadopoulos M, Gallo V, et al. Computational analysis of cortical neuronal excitotoxicity in a large animal model of neonatal brain injury. J Neurodev Disord. 2022;14:26.
- Fischer CL, Bates AM, Lanzel EA, Guthmiller JM, Johnson GK, Singh NK, Kumar A, Vidva R, et al. Computational models accurately predict multi-cell biomarker profiles in inflammation and cancer. Sci Rep. 2019;9:10877.
- Drusbosky LM, Vidva R, Gera S, Lakshminarayana AV, Shyamasundar VP, Agrawal AK, et al. Predicting response to BET inhibitors using computational modeling: a BEAT AML project study. Leuk Res. 2019;77:42-50.
- Sanidas G, et al. (incl. Vidva R). Prematurity insults remodel cerebellar development and behavior; Intrinsic gestational timing governs human cerebellar development after preterm birth. bioRxiv preprints, 2025, revised 2026.

Full list of 7 papers, 2 preprints, and 14 abstracts: orcid.org/0000-0002-2604-5448

Professional development: immunology course series (Imperial College London, 2023; Rice University, 2020 and 2021); Cancer Biology (JHU, 2018); Design and Interpretation of Clinical Trials (JHU, 2019); Genomic and Precision Medicine (UCSF, 2017); Drug Development Product Management (UCSD, 2017); Biotechnology and Intellectual Property (WIPO, 2012). ISCB member since 2024.

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